

The performance as of Feb 26 2025 of the latest model of big quiet fan (funded by grant A42)

Note Sept 22 2024: This document describes a slightly inferior older prototype. The latest prototype gets the same pressure/flow relationship, but with lower noise due to the use of a polymer with better damping. Noise is only about 42 dBa now.

This document describes the performance of the Big Quiet Fan, latest model as of Feb 25, the test procedures/data reliability, and comparisons to the next best fan for flow to noise ratio in the context of building air filter boxes, which is apparently the Arctic P14 (the SickelFlow is also pretty good but very similar so I focussed on the P14 since it's commonly used and is the next best fan discussed in the grant application). The Big Quiet Fan revision under test is the Version 11 primary version 10 secondary, version 6 frame, for reference.

It's important to note that work continues, to evolve the fan in several ways, by improving the motor driver to eliminate or reduce motor hum, and also to improve the blade design (probably only minor progress will be made on the blade design from this point on). The motor driver will be replaced with the open source one I am working on with the SimpleFOC community (especially Sequoia) at some point.

My old thermanemometer fried itself, so I got a new one and validated it using a rotating arm apparatus for the measurements in this document (pictures and brief description below). This one works better at low velocity, and it's accuracy is now known and can be corrected, even, using the table built during the validation process.

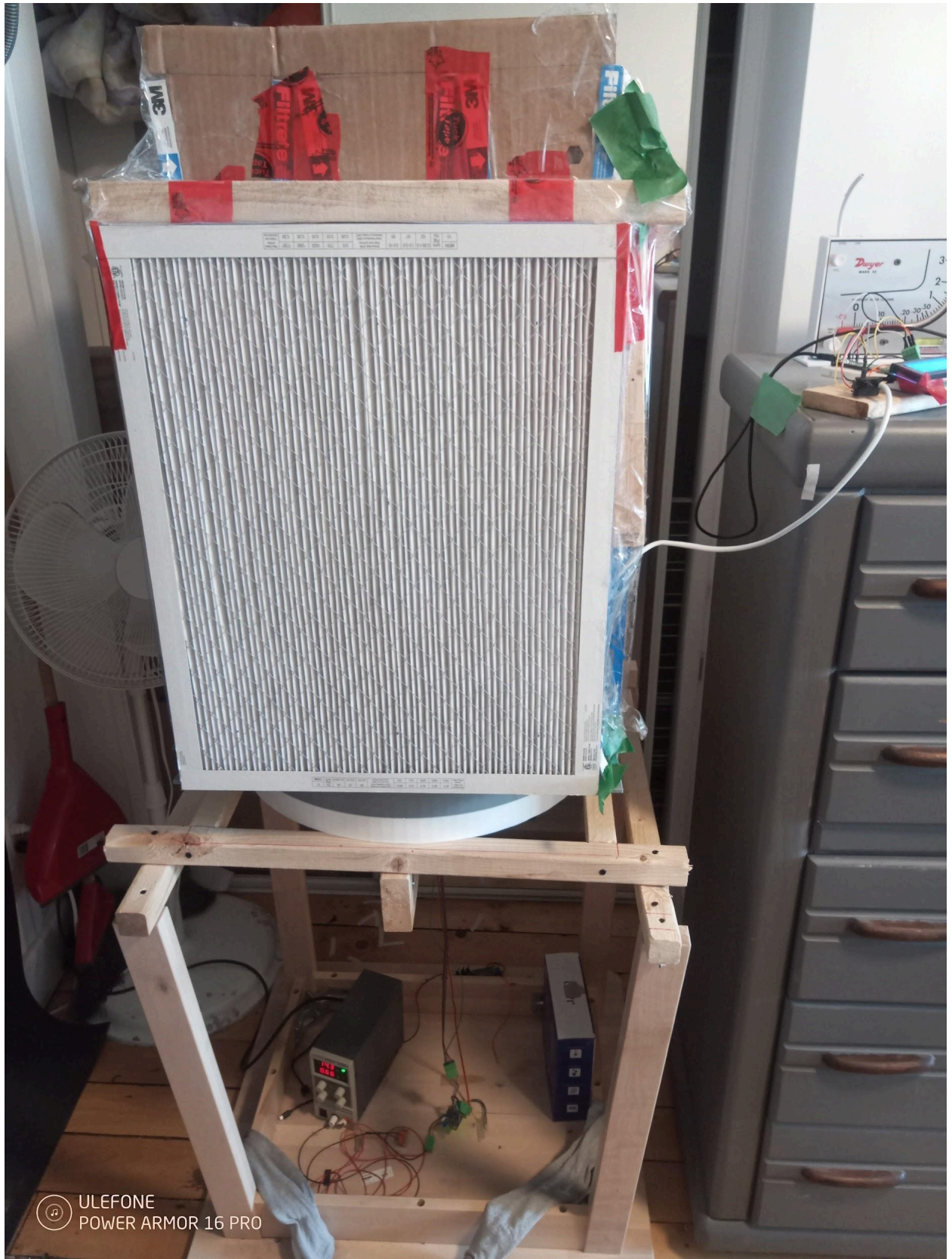
There are four sources of data I can use:

- ATOM, the engineering firm that Air Support Project hired to do various things, including evaluate fans and filters. That data is for the old V6 fan, which has inferior performance, and also there are some problems with the acoustic data, making it unusable (it's sound power rather than the standard sound pressure level at 1 meter, which can be converted crudely to SPL at 1 meter, but also there are considerable actual errors, as shown by my sound meter). The measurements of the flow to pressure ratio of the filters also does not match my measurements at all/cannot be replicated. This makes me trust the data very little. It was also taken at different RPM values than my standard test points, which make it hard to use precisely. I take data at 300 and 400 rpm usually, taking a range of data at a range of RPMs every time I changed something would not be practical or cost effective. For these reasons I have not used this data much, but we can use the flow/pressure values for the P14 fan at least, which I have also measured, and my measurements of those parameters are essentially in line with theirs, and also the manufacturer's (<https://support.arctic.de/products/p14-pwm-pst/techdocs/P14%20-%20PQ%20Curve%20for%20CFD.XLSX>) (the manufacturer's indicates slightly higher values, and at the same time lower rpms which would make it quieter).

- The second source is Joshua Pearce's lab, at the University of Western Ontario, who has taken thermal data mostly but is planning to get some acoustic data soon. This is the most reliable because they are academics that are in the habit of being quite precise to protect their reputation, but does not include pressure and flow data yet, and it's going to be a while before they get around to that. They mostly want it for a greenhouse anyway so such testing is not a priority. They have however already produced a report on some aspects of the fan, which IMO does not have much bearing here except to indicate it works ok and verify the power consumption at various RPM, but you can see it at https://docs.google.com/document/d/1Fr6ZwluvgL4xyAOdBe_uR3tBEltB9S_Q/edit?usp=sharing&oid=117287514477622889273&rtpof=true&sd=true
- The third is information on noise and subjective evaluation from HouseFresh, the organization in the UK that tests air filter appliances. I sent them a V11P/V10S fan mid january I think, shipping took a long time and they are getting around to testing soon hopefully. They have a class 2 sound meter and will build a filter box appliance of some kind and test the CADR with their usual methodology, which is unfortunately not a direct measure of absolute CADR but can be used as an apples to apples measure to compare the CADR of different appliances they have tested, they are still working on that, I have discussed this with them.
- Lastly, there is the data I have collected, which I have the most confidence in because I know exactly how it was collected, it is for the right fan and filter arrangement, it is at the right RPMs and using the same equipment and environment every time. Although done on a small budget, and I have learned the hard way that bias takes many forms, it is I think quite dependable at this point. This is because of the anemometer validation, and also the manometer, sound meter, ammeter, voltmeter (built into power supply), and tachometer are pretty solid.
 - The sound meter is an IEC class 2 meter which is a regulated device used for checking noise levels in the workplace for regulatory compliance, it's accuracy is 2-3 dB. It's precision/repeatability is considerably higher than that though, so the measurements I have made of the ratio of the noise produced by the P14 vs BQF for instance, largely circumvent that concern. The difference between two noise sources should be reflected with a precision and accuracy considerably higher than 2-3 dB. The background noise is the main source of error, it can theoretically be subtracted from the signal, and I have tried to do this, but the process can amplify error to some moderate degree. I do not know how the manufacturing chain of this device goes or where the regulatory oversight plays in but it appears to be reliable from every angle I have been able to look at it. Checking it the way I checked the anemometer is not something I can presently see a way to do. A rotating rod of precise geometry and rotational speed could be a reliable source of noise for calibration, but that's a pipe dream for now. I have checked it in several ways, for instance two identical noise sources is higher than one by 3 decibels, which is as expected. The noise level indicated also goes down with distance from the source as expected.

- The manometer is an SDP810-125A sensor, which I have used many times in various contexts and found remarkably reliable, and is consistent with water manometer and tilted oil manometer readings. I have a tilted oil manometer to verify things, but it's not as reliable, repeatable or accurate as the SDP810-125A. The data sheet says the SDP810-125A is calibrated at the factory and is accurate to within 0.1 pascals even at these low pressures. Ironically, it's actually a thermanemometer internally - pressure is inferred from the rate of flow through a small hole - but because the flow pattern etc. can be controlled inside the casing, it is consistent and reliable.
- The ammeter and volt meter are not particularly important but are built into the power supply. I have checked them incidentally many times over the years with a VOMM and it always seemed to agree.
- The tachometer is important for reliability but it's accuracy and repeatability come directly from electronic oscillators and electronics/optical devices so it's precision and accuracy is not something I am worried about.

Apparatus:







The apparatus consists of the filter set, which is 10 3M Filtrete brand 20 inch by 25 inch MERV 13 filters taped into an arrangement with a “triple V” in the middle, and 4 filters around the periphery. This is meant to be close to the final filter appliance I am optimizing for. In any optimality search there has to be a certain context, end use parameters of some kind, and I have adopted a set of 10 filters, in this arrangement, as that context.

There is some space internally for air to flow at the flat sides of the V, but air is still being restricted quite significantly in many areas, which could be improved by spacing the filters further apart in the final appliance. It provides the needed functionality for testing just fine, though. For example the total flow in the top of the device is only 2.5 times the flow rate of the filters on the side, despite there being 6 filters for air to flow through the triple V. Secondly, the variation in flow velocity between the top of the side filters and the bottom of the filters is about a factor of 3. This indicates very considerable effect of aerodynamic pressure drop internal to the assembly in the different areas, which I did not expect.

Again, this internal restriction of airflow was the main source of error in early measurements and extrapolation, combined with the fact I could not just measure the airflow because my anemometer did not function at these low flow values. The new one does.

It may not be clear from the picture, but the center timber is 1.5 inches away from the fan, allowing air to flow reasonably well although I'm sure it has a small negative impact, given the high flow rates and low pressures.

The fan was powered up to 300 and then 400 RPM, and the validated themanemometer used to measure the flow at the points indicated in the diagram in in the spreadsheet, and entered into the spreadsheet listed.

Spreadsheet of test data and computations:

<https://docs.google.com/spreadsheets/d/1RyS2FexTkQHimgX9ASrzR0OXF6ZEogR1g6wwQfYa-e4/edit?usp=sharing>

(See the other tabs for flow rate measurements of the actual test rig)

Conclusion of testing drawn from spreadsheet:

300 rpm: 37.5 dBa at 1 meter with filters on. 3.8 pascals, 1111 CFM.

400 rpm: 47.3 dBa at 1 meter with filters on. 5.6 pascals, 1452 CFM.

Edit march 12 2025: There should be certainty or error bars here, unfortunately they would themselves be estimates, and as you will see we can actually cancel out some of the latent error when comparing things.

Higher flow may be possible with a better filter arrangement.

Lower noise may be possible especially in the 300 rpm range with a motor that makes less noise, or modifications to the motor driver to cancel out the hum (anti-cogging), it is a considerable fraction of the

noise, I don't know exactly how much, but I can hear it. Also a spectrum analyser app indicates this to be true as well. This is in progress.

Comparison with P14 fan:

At 300 RPM:

At the pressure of 3.8 pascals (0.015265 inches water column), the P14 fan according to ATOM's data(<https://docs.google.com/spreadsheets/d/1a7mojU1EStVRD1M9MlaQoEpgtmqV-c7MnrezLvrl9aY/edit?usp=sharing>) gets about 62.5 CFM (halfway between the second and third data points).

The noise level of the P14 fan operating at 12 volts and indeed the same 1800 RPM, I measured with my sound meter is 35 decibels Sound Pressure Level at 1 meter (the ATOM data is Sound Power level, which is not the same measure, that's why it's different). This is done by subtracting the linearized background noise pressure level reading from the linearized noise level and then converting back to decibels (Assumed actual noise level emitted by device = $10 \cdot \log_{10}(10^{**}(\text{reading}/10) - 10^{**}(\text{background}/10)))$).

I confirmed the flow rate appears to be about right using my anemometer, a 6 inch diameter pipe and a plastic adapter plate, the pieces used are shown in the pictures below. Even a pipe of this diameter causes significant pressure drop at this flow rate, about 4 pascals. This kind of phenomena is the same that led to the internal flow restriction inside the filter arrangement.

The flow rate at this pressure of the BQF is therefore $1111/62.5 = 17.78$ times the flow rate of a single P14 fan. The noise level of the BQF is 2.5 decibels higher, therefore $10^{2.5/10} = 1.78$ times the noise level of a single P14 fan (sound pressure level is proportional to sound power emitted by a device, all else being equal). Therefore the flow to noise ratio is **9.98** times higher. **Edit march 12 2025:** Again, the question of error bar or uncertainty is important here, as relatively small variations in the dBa difference will affect this figure quite considerably. However, the main point is that it's pretty good. The input figures would have to change very considerably for this to drop below 1. In my opinion it's a very safe bet that things are quite good, when you sum it all together.

The original goal in the proposal was to get 16 times the flow of a P14 fan, at the relevant pressure, while getting the same or better flow to noise relationship as a P14 fan (1 or higher). With a ratio of 10 times higher flow to noise, I have therefore considerably exceeded the goals at this pressure and flow test point, which is the most relevant.

Again it is possible to reduce the noise of the BQF further by eliminating the motor hum.

At 400 RPM:

The BQF gives 5.6 pascals through the filter assembly at 400 RPM. At 5.6 pascals (0.0225 inches water column), the P14 gets 57 CFM, same noise level as at 3.8 pascals.

From the data above, the BQF therefore gets **25.45 times the flow** at this RPM and pressure, as a P14 fan. It gets 12.5 decibels higher noise, which is 17.78 times the noise. Therefore the ratio of flow to noise at this pressure and RPM is **1.43**. Clearly an array of 25 p14 fans is not practical, so this is a considerable advance, and greatly exceeds the original proposal's goal of 16x the flow and 1x the flow to noise ratio.

Confirmation of noise level from UWO and HouseFresh:

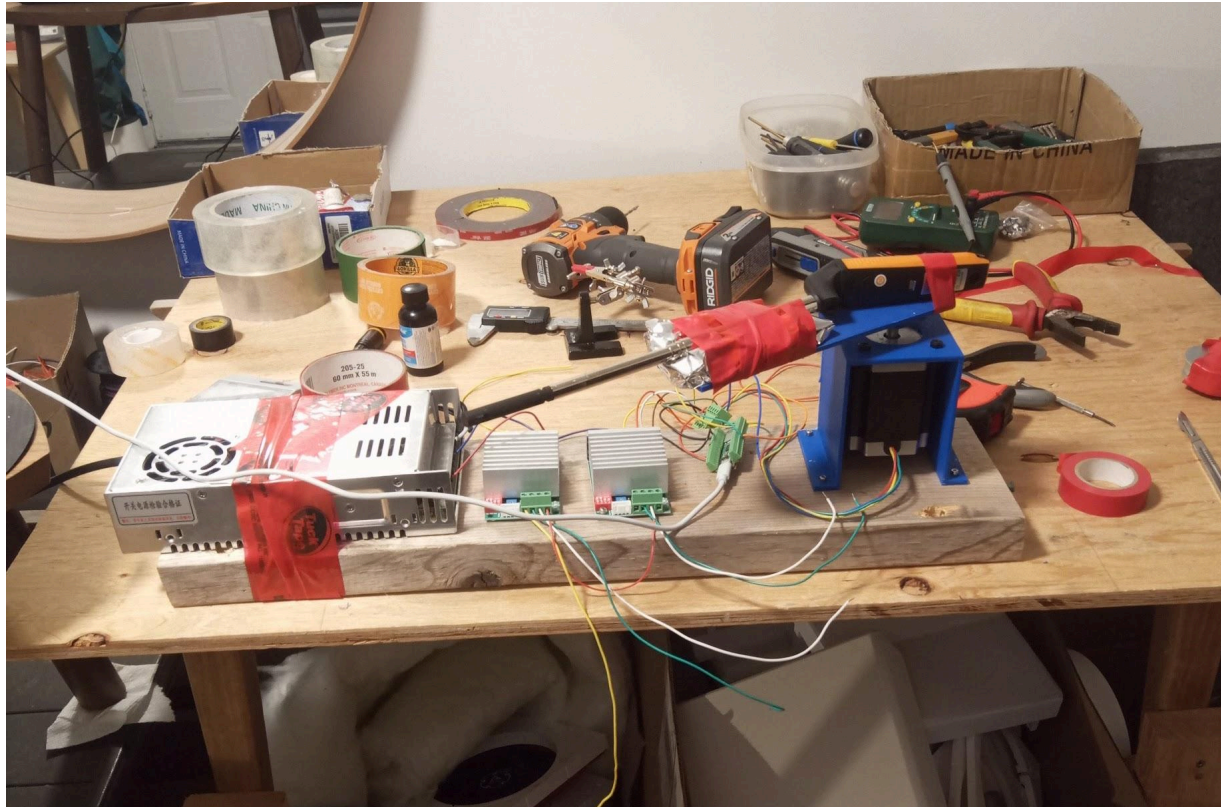
This is pending.

More pictures:









The anemometer test apparatus actually used a TMC2209 stepper driver to get smooth motion, the two stepper drivers in the foreground are not used. The distance between the center of rotation and the sensor of the thermanemometer is measured, and then the sensor rotated at a known speed. Data was taken and a spreadsheet constructed to ensure it was adequately reliable and allow for compensation to be done, even.